

$$X = \begin{bmatrix} 1 & 5 & 11 & 125 \\ 1 & 7 & 9 & 155 \\ 1 & 9 & 7 & 175 \\ 1 & 11 & 13 & 220 \\ 1 & 13 & 18 & 235 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 5 & 7 & 9 & 11 & 13 \\ 11 & 9 & 7 & 13 & 18 \\ 125 & 155 & 175 & 220 & 235 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 5 & 45 & 66 & 910 \\ 45 & 445 & 644 & 9220 \\ 66 & 644 & 920 & 13225 \\ 910 & 9220 & 13225 & 197400 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 22.2 \\ 58.2 \\ 22.2 \\ 58.2 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 6.503 & -0.4495 & -0.2329 & 0.0006 \\ -0.4495 & 0.0359 & 0.0110 & -0.0001 \\ -0.2329 & 0.0110 & 0.0177 & -0.0001 \\ 0.0006 & -0.0001 & -0.0001 & 0 \end{bmatrix}$$

$$Y = \begin{bmatrix} 48 \\ 72 \\ 101 \\ 142 \\ 176 \end{bmatrix}$$

Regression equation: $\hat{y} = 0.87x + 22.2$
 Coefficient for wing speed = 0.87
 Coefficient for block angle = 22.2
 Coefficient for water speed = 0.33
 Intercept = 22.2

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 5 & 7 & 9 & 11 & 13 \\ 11 & 9 & 15 & 13 & 18 \\ 125 & 155 & 175 & 220 & 235 \end{bmatrix} \begin{bmatrix} 48 \\ 72 \\ 101 \\ 142 \\ 176 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 539 \\ 5503 \\ 7705 \\ 107435 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 5 & 44 & 44 & 500 \\ 44 & 220 & 220 & 2660 \\ 44 & 220 & 220 & 2660 \\ 500 & 2660 & 2660 & 10000 \end{bmatrix}$$

$$\beta = (X^T X)^{-1} (X^T Y)$$

$$\beta = \begin{bmatrix} -55.4 \\ 6.87 \\ 6.55 \\ 0.37 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 2.0000 & -0.0222 & -0.0222 & 0.0000 \\ -0.0222 & 0.0022 & 0.0022 & -0.0002 \\ -0.0222 & 0.0022 & 0.0022 & -0.0002 \\ 0.0000 & -0.0002 & -0.0002 & 0.0000 \end{bmatrix}$$

$$\text{Regression equation, } y = 6.87x_1 + 6.55x_2 + 0.37x_3 - 55.4$$

$$\text{Coefficient for Wind speed} = 6.87$$

$$\text{Coefficient for Blade angle} = 6.55$$

$$\text{Coefficient for rotor speed} = 0.37$$

$$\text{Intercept} = \underline{\underline{-55.4}}$$

$$\text{Power Output for } (9, 11, 100) = \underline{\underline{28-208 \text{ kW}}}$$



